

# Mixed agreement vs. case assignment patterns

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## 1 Introduction

Agreement and case assignment seem to be two very different phenomena. When two constituents agree, they usually agree in all appropriate morphosyntactic features. For example, in many languages adjectives and nouns agree in all the features they share, e.g., in various Slavic languages, in number, gender, and case. By contrast, case assignment – whether it is conditioned by configurational, functional, lexical, or semantic factors (see Butt & King 2003, 2004, as well as King 1995: §8.1, and Butt 2023: §7, for summaries) – results in the exponence of a specific grammatical case, but it does not depend on the presence of another constituent bearing the same case value. In some syntactic constructions, both phenomena may be observed, as in many nominative-accusative languages, where subject-verb agreement in  $\phi$  features (i.e., number, gender, and – where applicable – person) is accompanied by the nominative case assignment to the subject.

However, there are constructions that display a more complex interplay of agreement and case assignment. One such construction, studied in various frameworks (see section 5 for literature overview), but not in Lexical Functional Grammar (LFG; Bresnan 1982, Bresnan et al. 2016, Dalrymple et al. 2019, Dalrymple 2023), is the numeral-noun construction (NNC) in North Slavic languages.<sup>1</sup> This is illustrated in table 1, which shows forms of two numeral phrases, differing in the gender of the noun, in various case positions: *pjat’stul’ev* ‘five chairs.M’ and *pjat’krovatej* ‘five beds.F’.<sup>2</sup>

Table 1: Russian numeral-noun constructions

| position | ‘five chairs’         |                          | ‘five beds’           |                            |
|----------|-----------------------|--------------------------|-----------------------|----------------------------|
|          | ‘five’                | ‘chairs.M’               | ‘five’                | ‘beds.F’                   |
| NOM      | <i>pjat’.PL.NOM</i>   | <i>stul’ev.PL.M.GEN</i>  | <i>pjat’.PL.NOM</i>   | <i>krovatej.PL.F.GEN</i>   |
| ACC      | <i>pjat’.PL.ACC</i>   | <i>stul’ev.PL.M.GEN</i>  | <i>pjat’.PL.ACC</i>   | <i>krovatej.PL.F.GEN</i>   |
| GEN      | <i>pjati.PL.GEN</i>   | <i>stul’ev.PL.M.GEN</i>  | <i>pjati.PL.GEN</i>   | <i>krovatej.PL.F.GEN</i>   |
| DAT      | <i>pjati.PL.DAT</i>   | <i>stuljam.PL.M.DAT</i>  | <i>pjati.PL.DAT</i>   | <i>krovatjam.PL.F.DAT</i>  |
| LOC      | <i>pjati.PL.LOC</i>   | <i>stuljax.PL.M.LOC</i>  | <i>pjati.PL.LOC</i>   | <i>krovatjax.PL.F.LOC</i>  |
| INS      | <i>pjat’ju.PL.INS</i> | <i>stuljami.PL.M.INS</i> | <i>pjat’ju.PL.INS</i> | <i>krovatjami.PL.F.INS</i> |

In nominative and accusative positions, the noun occurs in the genitive case. While both the numeral and the noun are plural (as can be verified by the forms of agreeing adjectives, not shown here), it is not

<sup>1</sup>Slavic languages are traditionally subdivided into East Slavic (e.g., Russian, Ukrainian), West Slavic (e.g., Czech, Polish), and South Slavic (e.g., Bulgarian, Croatian), with the less established term ‘North Slavic’ sometimes used to refer jointly to East and West varieties.

<sup>2</sup>All abbreviations used in this paper follow the Leipzig Glossing Rules, with the addition of COLL to mark collective numerals.

clear whether this is true agreement, as in this construction neither form can be put in the singular to test whether the other would follow suit. Similarly, in line with the robust generalization that Russian gender is neutralized in the plural, the form of the numeral does not depend on the gender of the noun: the two columns of table 1 labelled ‘five’ are identical. In short, only case assignment is evident in NOM and ACC rows.

By contrast, in dative, locative, and instrumental positions, the noun and the numeral agree at least in case: both occur in the case appropriate for the syntactic position of the NNC. That is, only agreement is evidenced in DAT, LOC, and INS rows. (In the case of GEN, the data is interpretable as either assignment or agreement – or both.)

In terms of Babby (1987), Russian NNCs display a homogeneous pattern in the oblique cases (GEN, DAT, LOC, INS) and a heterogeneous pattern in the direct cases (NOM, ACC). In our terms, table 1 illustrates a mixed agreement vs. case assignment pattern – a MACA pattern in short. The problem with MACA constructions is that they cannot be simply described as always involving agreement or as always involving case assignment or as always involving both; which of these phenomena is observed depends on the case position of the whole construction.

Polish presents additional complications. One that is ignored in this paper is that, in nominative positions, the numeral bears the accusative – see, e.g., Przepiórkowski (1999, 2004), as well as Przepiórkowski & Patejuk (2012) for an LFG analysis. The other complication is that, unlike in Russian, some agreement is clear even in the direct case positions, as illustrated in table 2 with case paradigms for the masculine *pięciu chłopców* ‘five.M boys.M’ and the feminine *pięć dziewczyn* ‘five.F girls.F’.<sup>3</sup>

Table 2: Polish numeral-noun constructions

| position | ‘five boys’               |                            | ‘five girls’              |                               |
|----------|---------------------------|----------------------------|---------------------------|-------------------------------|
|          | ‘five.M’                  | ‘boys.M’                   | ‘five.F’                  | ‘girls.F’                     |
| NOM      | <i>pięciu</i> .PL.M.ACC   | <i>chłopców</i> .PL.M.GEN  | <i>pięć</i> .PL.F.ACC     | <i>dziewczyn</i> .PL.F.GEN    |
| ACC      | <i>pięciu</i> .PL.M.ACC   | <i>chłopców</i> .PL.M.GEN  | <i>pięć</i> .PL.F.ACC     | <i>dziewczyn</i> .PL.F.GEN    |
| GEN      | <i>pięciu</i> .PL.M.GEN   | <i>chłopców</i> .PL.M.GEN  | <i>pięciu</i> .PL.F.GEN   | <i>dziewczyn</i> .PL.F.GEN    |
| DAT      | <i>pięciu</i> .PL.M.DAT   | <i>chłopcom</i> .PL.M.DAT  | <i>pięciu</i> .PL.F.DAT   | <i>dziewczynom</i> .PL.F.DAT  |
| LOC      | <i>pięciu</i> .PL.M.LOC   | <i>chłopcach</i> .PL.M.LOC | <i>pięciu</i> .PL.F.LOC   | <i>dziewczynach</i> .PL.F.LOC |
| INS      | <i>pięcioma</i> .PL.M.INS | <i>chłopcami</i> .PL.M.INS | <i>pięcioma</i> .PL.F.INS | <i>dziewczynami</i> .PL.F.INS |

As mentioned above, a given construction may display both case assignment and agreement in all shared features (as, again, in subject-verb constructions in nominative-accusative languages). However, what is observed in Polish MACA patterns in NOM and ACC rows is case assignment of the genitive accompanied by agreement in only some of the morphosyntactic features shared by the noun and the numeral: certainly gender, but apparently not case (with the status of number less clear).

The aim of this paper is to propose and contrast two analyses of North Slavic NNCs: derivational in section 2 and LFG in section 3. That is, this paper is in the spirit of King (1995), which proposed derivational and LFG accounts of syntactic aspects of information structure. While most of the data discussed in sections 2 and 3 is well known (with the exception of the ‘something interesting’ idiosyncrasy), section 4 presents new data concerning collective numerals, uniformly ignored by previous analyses

<sup>3</sup>In Polish structuralist linguistics, 5 gender values are assumed (Mańczak 1956), including – apart from feminine and neuter – three masculine: human (M1), animate (M2), and inanimate (M3). (The names of these genders are a little misleading, as many M2 nouns are inanimate and some are human.) The difference in the numeral forms illustrated in table 2 is between M1 and the other four genders. For this reason, a human masculine noun, *chłopiec* ‘boy’, is used in the Polish table. However, in the Russian table, using such an animate noun would reveal – and necessitate explaining – additional complications of Russian to do with the fact that masculine animate direct objects are in the genitive instead of the expected accusative. Hence the difference in lexical items used in tables 1 and 2.

(with the exception of Przepiórkowski 1999: §5.3.2.2) and – perhaps not coincidentally – at first sight incompatible with those analyses. I will show that the usually expressed generalizations concerning Polish collective numerals are wrong and that the actual empirical facts are amenable to the Minimalist (Chomsky 1995b, 2000) analysis of section 2, as well as the LFG analysis of section 3. In section 5, the proposed analyses are briefly compared to previous work, and section 6 concludes.

## 2 Minimalist analysis

### 2.1 Main ideas

The Minimalist analysis of NNCs sketched here, presented in much more detail in Przepiórkowski (2025), is based on the following theoretical ideas, which are relatively framework-neutral in the sense that they may in principle be formalized in various frameworks, including LFG, as shown in section 3 below.

First, regardless of the syntactic position or the case assigned to that position, NNCs always involve both agreement, as if numerals were adjectives, and the assignment of the genitive case to the noun, as if numerals were themselves nouns. This is justified diachronically, given that Slavic numerals evolved from adjectives (low – i.e., paucal – numerals) and from nouns (high numerals), and that they share various properties of these two classes to various degrees (see, e.g., Corbett 1978 and much subsequent literature). This is also in line with the behaviour of Polish NNCs, where both overt gender agreement and genitive assignment are witnessed in the same (direct) positions.

However, I assume that relevant numerals and nouns agree fully, not only in gender, but also in number and, importantly, case. This means that nouns may bear two case values in NNCs: that resulting from agreement and the assigned genitive. For example, the overtly genitive noun *dziewczyn* ‘girls’ in the ACC row of table 2 actually bears both GEN and ACC, and the overtly instrumental form *dziewczynami* in the INS row actually bears both GEN and INS. In the Minimalist literature, this situation is sometimes referred to as *covert case stacking* (Richards 2013, Pesetsky 2013).

Second, which of the covertly stacked cases is overtly expounded is regulated by explicit exponence principles. I adopt here the rule assumed in Matushansky (2008) (cf. Ionin & Matushansky 2018: 185), namely: *expose the most marked case*.<sup>4</sup> Matushansky (2008) proposes this rule on the basis of case assigned to nominal predicates in Finnish and additionally justifies it with Russian NNCs, foreshadowing crucial aspects of the analysis proposed here. Additionally, Grishin (2024) applies it to Gothic free relatives.

Third, case markedness is not stipulated but it is derived on the basis of systematic syncretisms and morphological containments (e.g., Caha 2009, Smith et al. 2019; cf. Blake 2001). In particular, Caha (2009) argues at length, on the basis of phenomena independent of those discussed in the current paper, for the following case hierarchy for Slavic (and universally):<sup>5</sup>

- (1) NOM < ACC < GEN < LOC < DAT < INS (Caha 2009)

Note that the first idea (numerals having properties of both nouns and adjectives) directly reflects empirical observations and diachronic developments, while the other two ideas (the *expose the most marked* rule and the specific case markedness hierarchy in (1)) are motivated independently of the phenomena at hand. Together, they immediately account for MACA patterns observed in tables 1 and 2: given that GEN is more marked than NOM and ACC, it is expounded on the noun in nominative and accusative positions; by contrast, DAT, LOC, and INS are more marked than GEN, so they are expounded in oblique positions.<sup>6</sup>

<sup>4</sup>See Przepiórkowski (2025) for refinements necessitated by Russian paucals and Estonian pseudopartitives.

<sup>5</sup>As is usual, I ignore the vocative, as its case status is not clear (cf. Blake 2001: §1.2.3).

<sup>6</sup>One of the reviewers wonders how this analysis could be extended to Russian paucal numerals, which display a considerably more complex pattern; such an extension, but only in Minimalist terms, is proposed in Przepiórkowski (2025).

## 2.2 Technicalities

### 2.2.1 Concord

The account of NNCs proposed in Przepiórkowski (2025) is based on a novel approach to nominal concord. Current Minimalist theories of concord, such as Norris (2014, 2018) or Bayırlı (2017), postulate a number of feature percolation operations specific to concord – upward percolation of some features, e.g.,  $\phi$ , and downward percolation of others, e.g., case – which goes against the Minimalist aim to avoid such specialized operations. By contrast, the operation responsible for concord in Przepiórkowski (2025) is Agree, which is also responsible for other instances of agreement. A recent approach to Agree is assumed (Deal 2015, 2024, Clem & Deal 2024), on which a single probe may Agree with multiple accessible goals (so-called *insatiable Agree*). A possible side effect of Agree is flagging, i.e., marking goals with a specific case or with a specific feature bundle.

The schematic tree in (2), which corresponds to the bracketed nominal phrase in (3), illustrates this account of concord.<sup>7</sup>

- (2)
- 
- (3) Oddamy je [jakieś młodszemu dziecku].  
 give.FUT.1PL them.ACC.PL some.DAT.SG.N younger.DAT.SG.N child.DAT.SG.N  
 ‘We’ll give them to some younger child.’

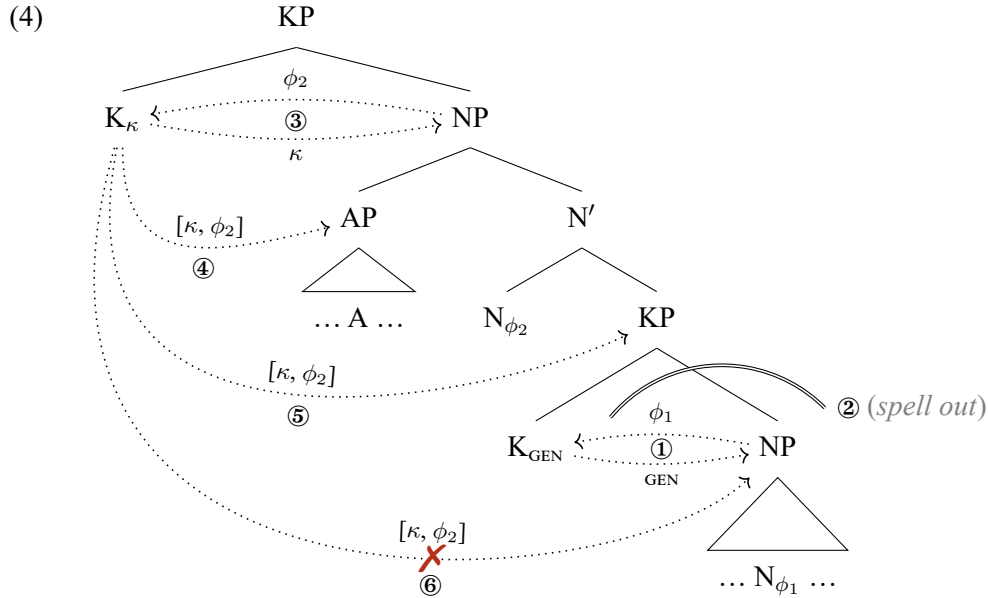
For simplicity, I assume that there are no DPs in Polish, only NPs, but nothing hinges on this assumption. Importantly, maximal nominal projections are assumed to be KPs, with the functional case head K phonologically empty in Polish, but possibly overt in some other languages (Bittner 1994, Bittner & Hale 1996a,b). Assuming a realizational approach to morphology (such as Distributed Morphology, Halle & Marantz 1993, or Nanosyntax, Caha 2009), some of the tree leaves are lexical roots which initially bear only some morphosyntactic features, those which are specific for the root (e.g., the grammatical gender of the noun) or which reflect the intended semantics (e.g., the singular number of the noun). Thus, the nominal leaf N in (2) is specified for number and gender ( $\phi$  in (2) = SG.N in (3)), but not for case, while the adjectival leaves A are initially not specified for any morphosyntactic features.

Assuming Bare Phrase Structure (Chomsky 1995a), N, N', and NP in (2) are actually the same feature bundle, i.e., they share all features, including  $\phi$ , in a way analogous to c-structure nodes along a spine sharing the same f-structure in LFG. In the first interesting step, ①, the insatiable probe on K Agrees with the NP, i.e., it copies over the  $\phi$  features of this nominal complement and, in the process, flags the NP with its case,  $\kappa$  (= DAT in (3)). Following Clem & Deal (2024), any subsequent probing amounts to flagging with the whole  $[\kappa, \phi]$  bundle; see steps ② and ③. This way not only the particular case  $\kappa$ , but also the noun's  $\phi$  features end up on all accessible constituents within the maximal nominal projection –

<sup>7</sup>Unless indicated otherwise, all examples come from the National Corpus of Polish (NCP; <https://nkjp.pl/>; Przepiórkowski et al. 2011, 2012).

in this case, on both adjectival phrases – creating the impression of direct agreement between these APs and the head noun.

The notion of accessible constituents refers to the standard Minimalist approach to locality, on which certain kinds of phrases are spelled out at certain points of derivation and, thus, become inaccessible for any further operations, including Agree (Chomsky 2000). In particular, maximal nominal projections – here, KPs – are assumed to be so-called phases, which means that NP complements of Ks are spelled out as soon as KPs are built. This prevents flagging of such NPs by higher insatiable probes, as illustrated with the tree in (4), which corresponds to example (5).



- (5) Pozwolił się uwieść [gładkiemu ciałku szefowej].  
 allowed.PST.3SG.M REFL seduce.INF smooth.DAT.SG.N body.DAT.SG.N boss.GEN.SG.F  
 ‘He allowed himself to be seduced by his boss’s smooth body.’

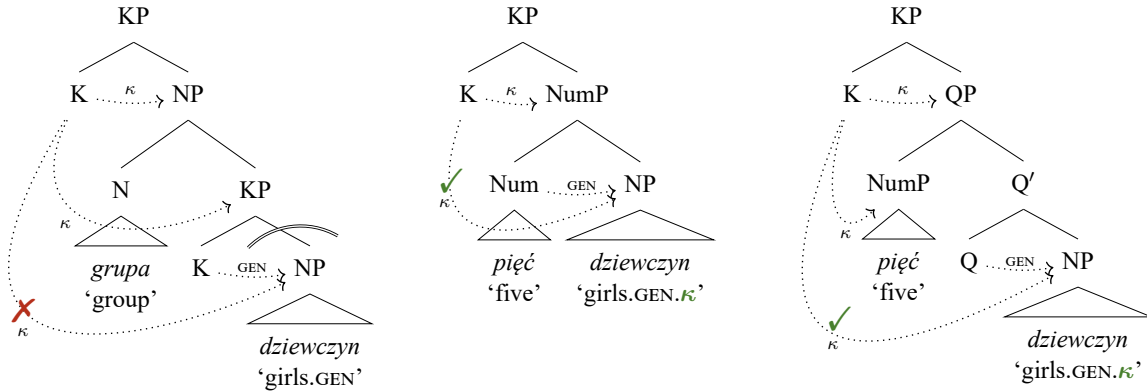
In the first interesting step ①, the lower GEN-assigning K combines with the lower NP, resulting in the genitive KP *szefowej* ‘boss.GEN.SG.F’. At this point, the lower NP is spelled out in step ②, so no further process can affect the form of this genitive constituent. Steps ③–④ in (4) are fully analogous to steps ①–③ in (2). In step ⑤, the lower KP is successfully flagged with  $[\kappa, \phi_2]$ , but features on K are not overtly expored in Polish, so this has no visible effect, in contrast to overt case stacking languages such as Kardil (see Przepiórkowski 2025 for details). Finally, an attempt at flagging in ⑥ fails, as the lower NP has already been spelled out at this stage.

### 2.2.2 Numeral-noun construction

There is no agreement in derivational literature on the structure of Slavic NNCs. On some approaches (e.g., Klockmann 2017), NNCs are headed by the numeral, as schematically illustrated in (7) for the NNC *pięć dziewczyn* ‘five girls’. On other approaches (e.g., Witkoś et al. 2018), numeral phrases are in the specifier position of an empty Q head, as in (8). However, in all approaches I am aware of, the noun belongs to the same extended projection (Grimshaw 2005) as the numeral. This is in contrast to nominal phrases containing embedded genitive nominal phrases, e.g., *grupa dziewczyn* ‘a group of girls’ in (6), where the embedded genitive nominal phrase is a separate KP, the crucial parts of which are spelled out before the higher K merges, as discussed above. Since no separate KP encompasses the noun in NNCs, it may be flagged by the K with the case  $\kappa$ , in addition to receiving the genitive from the numeral (as in (7)), from the empty Q (as in (8)), or by some other process specific to a given analysis of NNCs. Hence, whatever the exact source of the genitive, the noun ends up with two – covertly stacked – cases:

GEN and  $\kappa$ . Given the *expose the most marked* principle and Caha’s (2009) case hierarchy in (1) above, the NNC facts in tables 1 and 2 follow: when  $\kappa \in \{\text{NOM}, \text{ACC}\}$ , the markedness competition is won by GEN, when  $\kappa = \text{GEN}$ , the two competing cases are both GEN, so GEN is exposed, and finally when  $\kappa \in \{\text{DAT}, \text{LOC}, \text{INS}\}$ , the competition is won by  $\kappa$ , as it is more marked than GEN.

(6) noun-noun construction: (7) NNC (Num is head): (8) NNC (NumP is in Spec,QP):



On this analysis, there is nothing special about numeral or Q heads that results in MACA patterns, apart from the universally assumed property that they belong to the same extended projection as the noun and that they are able to assign GEN, perhaps by virtue of being (partially) functional heads (see Klockmann 2017 on semi-lexicality). This means that similar patterns might occur outside the domain of numeral phrases. This prediction is borne out by Estonian pseudopartitives discussed in Norris (2014, 2018) and analysed in the current framework in Przepiórkowski (2025), but also by a certain rarely discussed quirk of Polish, to which we turn now.

### 2.3 Something interesting

There are a handful of Polish pronouns, all containing the root *c(o)*, which display the MACA pattern similar to that of numerals, including: *co* ‘what’, *coś* ‘something’, *cokolwiek* ‘whatever’, and *nic* ‘nothing’; see table 3. Note that, while in NNCs the numeral occurs in the expected case and the noun displays the mixed case pattern, here the pronoun occurs in the expected case and it is the adjective that displays the mixed pattern.

Table 3: Something (and somebody) interesting in Polish

| position | ‘something interesting’ |                       | ‘somebody interesting’ |                       |
|----------|-------------------------|-----------------------|------------------------|-----------------------|
|          | ‘something.N’           | ‘interesting.N’       | ‘somebody.M’           | ‘interesting.M’       |
| NOM      | <i>coś</i> .NOM         | <i>ciekawego</i> .GEN | <i>ktoś</i> .NOM       | <i>ciekaw</i> y.NOM   |
| ACC      | <i>coś</i> .ACC         | <i>ciekawego</i> .GEN | <i>kogoś</i> .ACC      | <i>ciekawego</i> .ACC |
| GEN      | <i>czegoś</i> .GEN      | <i>ciekawego</i> .GEN | <i>kogoś</i> .GEN      | <i>ciekawego</i> .GEN |
| DAT      | <i>czemuś</i> .DAT      | <i>ciekawemu</i> .DAT | <i>komuś</i> .DAT      | <i>ciekawemu</i> .DAT |
| LOC      | <i>czymś</i> .LOC       | <i>ciekawym</i> .LOC  | <i>kimś</i> .LOC       | <i>ciekawym</i> .LOC  |
| INS      | <i>czymś</i> .INS       | <i>ciekawym</i> .INS  | <i>kimś</i> .INS       | <i>ciekawym</i> .INS  |

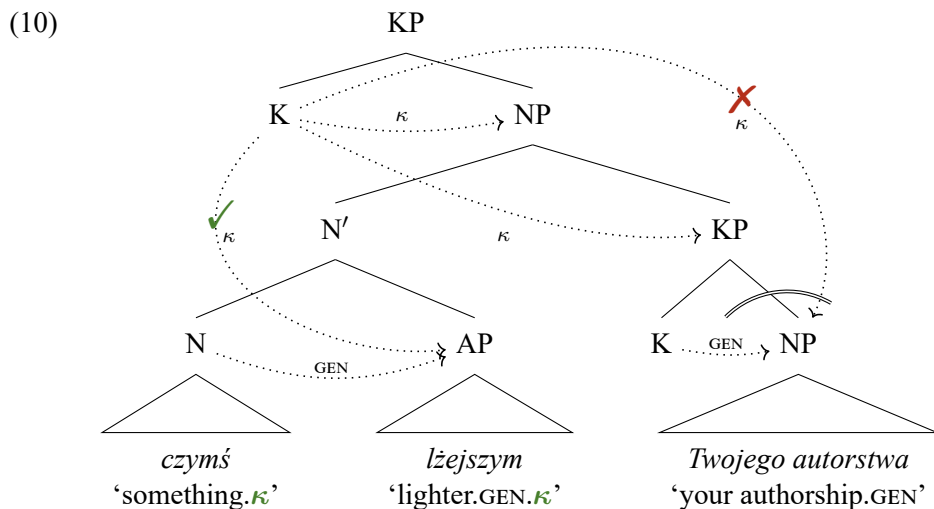
This behaviour should be contrasted with the usual situation, exemplified in the final columns of table 3, which illustrate that pronouns based on the root *kt(o)*, e.g., *ktoś* ‘somebody’, do not display the MACA pattern; here the adjective always agrees with the pronoun, as it does elsewhere in Polish.

It is not clear why exactly *coś* ‘something’, but not *ktoś* ‘somebody’, flags its adjectival modifier with GEN.<sup>8</sup> But, given that it does, the pattern shown in table 3 is fully expected on the approach to concord presented above: since AP is not (within) a separate phase, the higher K may flag this AP with case  $\kappa$ , on top of GEN received from *coś*, resulting in covert case stacking and the realization of the most marked case. On this account, the only idiosyncrasy of the Polish pronominal root *c(o)* is that it flags the adjective with GEN; the mixed pattern seen in table 3 follows from the lack of phase boundary between the higher K and the AP and from the *expose the most marked* principle.

This is confirmed by the fact that – in contrast to such AP dependents – genitive KP dependents of *coś* ‘something’ must stay in the genitive regardless of the external case, as illustrated with the constructed example (9) (based on a similar corpus example).

- (9) Porozmawiajmy o [czymś lżejszym Twojego autorstwa].  
 talk.IMP.1PL about something.LOC lighter.LOC your.GEN authorship.GEN  
 ‘Let’s talk about something lighter (written) by you.’  
 (lit.: ‘...something.LOC lighter.LOC [your authorship].GEN’)

The structure of such nominal phrases is schematically given in (10) (where  $\kappa = \text{LOC}$  in (9)), without commitment to any particular position of the AP and the inner KP; what is crucial is only that the inner KP is a phase, so the NP within it is not accessible to the probe on the higher K, while there is no similar phase boundary between the higher K and the AP.



In summary, the ‘something interesting’ idiosyncrasy is in line with the general approach to concord and the particular analysis of Slavic numeral-noun constructions offered above. There is nothing special about numeral constructions that gives rise to the famous MACA pattern. This conclusion is additionally confirmed by the observation in Rutkowski & Szczegot (2001) that some Lithuanian pronouns, including *kažkas* ‘something’, behave similarly to Polish *coś* ‘something’, even though Lithuanian NNCs do not display the Slavic MACA pattern.

Having presented a Minimalist formalization of the relatively framework-independent ideas sketched in section 2.1, let me now propose a corresponding LFG analysis.

<sup>8</sup>In terms of Klockmann (2017), the root *c(o)* – but perhaps not *kt(o)*, etc. – would be analysed as idiosyncratically semi-lexical, in the sense that it hosts certain functional features (here; a GEN-flagging probe); see also Rutkowski & Szczegot (2001) for a similar analysis.

### 3 LFG analysis

The following analysis is formulated in standard LFG terms, but it is inspired by some aspects of L<sub>R</sub>FG (Melchin et al. 2020, Asudeh & Siddiqi 2022, 2024, Asudeh et al. 2023, 2024) – a still evolving version of LFG which does not assume the Lexical Integrity Principle (Bresnan & Mchombo 1995, Asudeh et al. 2013), but instead relies on the “syntax all the way down” approach of Distributed Morphology (Halle & Marantz 1993).

#### 3.1 Main ideas

The main ideas of the LFG analysis are essentially the same as in the case of the Minimalist analysis presented in section 2 above.

First, NNCs involve both concord and the assignment of GEN to the noun.

Second, the realization of a specific case on the noun is regulated by explicit exponence rules, which rely on case markedness.

Third, the North Slavic case markedness hierarchy repeated below in (11) is not stipulated but reflects morphological syncretism and containment facts discussed in Caha (2009).

$$(11) \quad \text{NOM} < \text{ACC} < \text{GEN} < \text{LOC} < \text{DAT} < \text{INS} \quad (\text{Caha 2009})$$

#### 3.2 Technicalities

##### 3.2.1 Concord

In this subsection, I introduce some assumptions and an initial version of the proposed approach to concord, which will be slightly modified in the following section 3.2.2.

Technically, I follow Asudeh et al. (2024) and Belyaev et al. (2025) in representing cases as feature matrices, with features such as NOM, ACC, or DAT understood as privative: either present (with the + value) or absent.<sup>9</sup> Case hierarchies such as (11) – but also more complex, non-linear hierarchies – are encoded via macros (also called *templates*; Dalrymple et al. 2004, Asudeh et al. 2013) in such a way that macros corresponding to more marked cases in the hierarchy call macros corresponding to immediately less marked cases. The six macros defined in (12)–(17) are needed in the case of (11).<sup>10</sup>

$$\begin{aligned}
 (12) \quad @\text{NOM} &:= (\downarrow \text{CASE NOM}) = + & (18) \quad \begin{bmatrix} \text{PRED} & \text{'GIRL'} \\ \text{CASE} & \begin{bmatrix} \text{NOM} & + \\ \text{ACC} & + \\ \text{GEN} & + \end{bmatrix} \\ \text{PHI} & \begin{bmatrix} \text{NUMBER} & \text{PL} \\ \text{GENDER} & \text{F} \end{bmatrix} \end{bmatrix} \\
 (13) \quad @\text{ACC} &:= @\text{NOM} & (19) \quad \begin{bmatrix} \text{PRED} & \text{'AMBITIOUS'} \\ \text{CASE} & \begin{bmatrix} \text{NOM} & + \\ \text{ACC} & + \\ \text{GEN} & + \end{bmatrix} \\ \text{PHI} & \begin{bmatrix} \text{NUMBER} & \text{PL} \\ \text{GENDER} & \text{F} \end{bmatrix} \end{bmatrix} \\
 (14) \quad @\text{GEN} &:= @\text{ACC} \\
 (15) \quad @\text{LOC} &:= @\text{GEN} \\
 (16) \quad @\text{DAT} &:= @\text{LOC} \\
 (17) \quad @\text{INS} &:= @\text{DAT}
 \end{aligned}$$

While similar hierarchies are often assumed for particular  $\phi$  features (cf. the *feature geometry* of Harley & Ritter 2002), also in L<sub>R</sub>FG, I will not do that here, as a careful investigation of the Polish gender

<sup>9</sup>See Dalrymple et al. (2009) for a similar but distinct idea: there, such attributes are binary (valued + or –, or absent) and they are used to model syntactic effects of syncretism.

<sup>10</sup>These macros are defined in such a way that they can be used in syntactic rules, e.g., to mark nominal dependents of numerals as genitive, as in (26) in the main text below. More general – parameterized – macros are needed to be also useful in lexical entries, e.g., to specify a given argument of a verb as genitive; such parameterized macros are introduced in section 3.3.

system, with its 3 masculine genders (cf. footnote 3), deserves a separate treatment. However, I will group the  $\phi$  features into a separate feature matrix (cf. INDEX in HPSG), so that the basic f-structure of the form *dziewczyn* ‘girl.GEN.PL.F’ is as in (18) above.

I moreover follow Haug & Nikitina (2015) in adopting the “feature sharing” approach to agreement, with agreement features represented directly on all agreeing constituents (cf. Haug 2023).<sup>11</sup> For example, the syncretic adjective *ambitnych* ‘ambitious’, when interpreted as GEN.PL.F, is represented at f-structure as plural not just by virtue of this adjective modifying a plural noun, as in standard LFG, but by virtue of expressing this feature directly, as in (19).

Given these assumptions, fully symmetrical concord may be encoded via the macros in (20)–(22), used in rules as shown in (23), and leading to functional representations as in (24) (with variables [1] and [2] indicating structure sharing).

$$(20) \quad @CONCORD := @CASECONCORD \\ @PHIAGREEMENT$$

$$(21) \quad (\text{preliminary version, to be replaced with (25)}) \\ @CASECONCORD := (\uparrow CASE) = (\downarrow CASE)$$

$$(22) \quad @PHIAGREEMENT := (\uparrow PHI) = (\downarrow PHI)$$

$$(23) \quad (\text{preliminary version, to be replaced with (35)}) \\ NP \rightarrow AP^* \quad N \\ \downarrow \in (\uparrow ADJ) \quad \uparrow = \downarrow \\ @CONCORD$$

$$(24) \quad \textit{ambitnych dziewczyn} \\ \text{‘ambitious.GEN.PL.F girls.GEN.PL.F’}$$

$$\left[ \begin{array}{l} \text{PRED ‘GIRL’} \\ \text{CASE [1] } \left[ \begin{array}{l} \text{NOM +} \\ \text{ACC +} \\ \text{GEN +} \end{array} \right] \\ \text{PHI [2] } \left[ \begin{array}{l} \text{NUMBER PL} \\ \text{GENDER F} \end{array} \right] \\ \text{ADJ } \left\{ \left[ \begin{array}{l} \text{PRED ‘AMBITIOUS’} \\ \text{CASE [1]} \\ \text{PHI [2]} \end{array} \right] \right\} \end{array} \right]$$

### 3.2.2 Numeral-noun construction

The gist of the analysis of Slavic NNCs is that two constituents may agree in  $\phi$  features (number, gender, and – where applicable – person) and case, but one of them may be additionally assigned a specific case, which sometimes results in case inequality – lack of case concord – between the two constituents. I implement this paradoxical “ $\phi$  and case concord but perhaps no case concord in the end” approach by modelling case concord as subsumption rather than equality, i.e., by replacing (21) above with (25) below.

$$(25) \quad @CASECONCORD := (\uparrow CASE) \sqsubseteq (\downarrow CASE) \quad (\text{final version, replaces (21)})$$

Note that nothing substantial changes in the analysis of simple adjective-noun concord illustrated with (23)–(24) above: the values of the two PHI features are still identical, and also CASE features are the same, but without the constraint that they must remain identical, so that some other part of the grammar may in principle further specify the value of the adjective’s CASE. As this does not happen in adjective-noun concord, the two CASE values remain identical.

However, this symmetry is broken in NNCs, which are licensed by syntactic rules such as (26). (Num stands for Numeral.)

$$(26) \quad \text{NumP} \rightarrow \text{Num} \quad \text{NP} \\ \uparrow = \downarrow \quad \downarrow = (\uparrow \text{OBJ}) \\ @CONCORD \\ @GEN$$

Such rules are directly responsible for MACA patterns. Let me illustrate this effect with numeral phrases in accusative and locative positions.

<sup>11</sup> This approach is typical of HPSG (Pollard & Sag 1994, Wechsler & Zlatić 2003), but it is also assumed in the derivational approach of Ackema & Neeleman (2013).

Assume a numeral phrase in an accusative position, i.e., with CASE value equal to [NOM+, ACC+]. Given the @CONCORD annotation in (26), this CASE value subsumes the CASE value of the noun, in accordance with the definition of @CONCORD in (25). This means that N's CASE has at least these two features: NOM+, ACC+. However, the @GEN annotation defines three CASE features on N: NOM+, ACC+, GEN+. Note that these two macros, @CONCORD and @GEN are compatible and result in the noun being specified as genitive. However, crucially, since the numeral's CASE only *subsumes* the noun's CASE rather than being *equal* to it, the numeral stays in the accusative, as illustrated in (27).

Assume now a NumP in a locative position, where the value of CASE is [NOM+, ACC+, GEN+, LOC+]. This value subsumes N's CASE, so the noun must also be “at least locative”. This time the presence of @GEN does not add anything to this specification, as the noun already contains the three features defining the genitive: NOM+, ACC+, GEN+. The resulting f-structure is illustrated in (28).

(27) *pięć dziewczyn* ‘five.ACC girls.GEN’

|   |          |             |         |
|---|----------|-------------|---------|
| [ | PRED ‘5’ |             | ]       |
|   | CASE     | [NOM +]     |         |
|   |          | [ACC +]     |         |
|   | PHI      | [1]         |         |
|   |          | [GENDER F]  |         |
| [ | OBJ      | PRED ‘GIRL’ |         |
|   |          | CASE        | [NOM +] |
|   |          |             | [ACC +] |
|   |          |             | [GEN +] |
|   | PHI      | [1]         |         |
|   |          |             |         |
|   |          |             |         |

(28) *pięciu dziewczynach* ‘five.LOC girls.LOC’

|               |             |           |
|---------------|-------------|-----------|
| PRED '5'      |             |           |
| CASE          | [ NOM + ]   |           |
|               | ACC +       |           |
|               | GEN +       |           |
|               | LOC +       |           |
| PHI           | [ 1 ]       |           |
| [ NUMBER PL ] |             |           |
| [ GENDER F ]  |             |           |
| OBJ           | PRED 'GIRL' |           |
|               | CASE        | [ NOM + ] |
|               |             | ACC +     |
|               |             | GEN +     |
| LOC +         |             |           |
| PHI           | [ 1 ]       |           |

In summary, the presence of both macros, @CONCORD and @GEN, on a given constituent in a syntactic rule is responsible for the MACA pattern discussed in this paper. The constituent will invariably be in the genitive when it agrees with a nominative, accusative, or genitive constituent, and it will occur in the agreeing case in locative, dative, and instrumental. As shown in the following section, this behaviour is not specific to NNCs and may be equally easily accounted for in ‘something interesting’ constructions. But a couple of technical comments are needed before moving there.

First, on the encoding of case assumed above, lexical entries cannot *define* case, they can only *expose* case defined in the grammar. For example, imagine that the LOC.PL.F form of *pięciu* ‘five’ had the lexical entry in (29). Then such a lexical entry would be compatible with the grammatical specification of case as accusative, i.e., as [NOM+, ACC+], which would result in an f-structure in (28), where case is locative, for the accusative numeral phrase *pięć dziewczyn* ‘five.ACC girls.GEN’.

(29) (bad lexical entry)

|               |     |                     |
|---------------|-----|---------------------|
| <i>pięciu</i> | Num | (↑ PRED) = ‘5’      |
|               |     | (↑ PHI NUMBER) = PL |
|               |     | (↑ PHI GENDER) = F  |
|               |     | (↑ CASE NOM) = +    |
|               |     | (↑ CASE ACC) = +    |
|               |     | (↑ CASE GEN) = +    |
|               |     | (↑ CASE LOC) = +    |

(30) (good lexical entry)

|               |     |                     |
|---------------|-----|---------------------|
| <i>pięciu</i> | Num | (↑ PRED) = ‘5’      |
|               |     | (↑ PHI NUMBER) = PL |
|               |     | (↑ PHI GENDER) = F  |
|               |     | (↑ CASE GEN)        |
|               |     | ¬(↑ CASE INS)       |

Instead, lexical entries should specify case via constraining statements, as in (30). This lexical entry is also more general than (29) was meant to be, as it defines a feminine form that exposes genitive,

locative, and dative (see table 2 for a fuller paradigm). This is done by ensuring the presence of CASE GEN, which excludes nominative and accusative, and denying the presence of CASE INS, which excludes instrumental.<sup>12</sup>

The second remark is that, as things stand at the moment, the analysis may be further simplified by assuming that *all* agreement is defined via subsumption rather than via identity. A single attribute, e.g., AGR, could collect all agreement features (NUMBER, GENDER, and particular case features such as ACC and LOC), and @CONCORD could be defined simply as ( $\uparrow$  AGR)  $\sqsubseteq$  ( $\downarrow$  AGR). However, exploring this possibility and its compatibility with alternative approaches to the representation of number and gender (e.g., as sets of privative features, as is common in L<sub>R</sub>FG) would take us too far afield, so only the more conservative solution – where  $\phi$  agreement is encoded as symmetric feature sharing – is considered here.

### 3.3 Something interesting

The Minimalist analysis of the ‘something interesting’ idiosyncrasy offered in section 2.3 is noncommittal about the exact position of the adjective in the NP, although the tree in (10) suggested that it is a complement of N rather than an ordinary modifier. The LFG analysis offered in this subsection makes this aspect of the analysis more explicit.

The crucial observation is that in Polish – just as in English – the AP in this construction is necessarily postnominal, as illustrated in the following constructed examples:

- (31) { [Coś ciekawego] / ?\*[Ciekawego coś] } się wydarzyło.  
 something.NOM interesting.GEN interesting.GEN something.NOM REFL happen.PST.3SG  
 ‘{[Something interesting] / \*[Interesting something]} happened.’
- (32) Porozmawiajmy o { [czymś interesującym] / ?\*[interesującym czymś] }.  
 talk.IMP.1PL about something.LOC interesting.LOC interesting.LOC something.LOC  
 ‘Let’s talk about {[something interesting] / \*[interesting something]}.’

Such postnominal adjectives have a special status in Polish (see, e.g., Rutkowski & Progovac 2005 and Rutkowski 2007, as well as references therein): unlike prenominal adjectives, they tend to refer to essential, not contingent, properties, or to particular subclasses of entities denoted by the noun. This difference is best illustrated with an example from Rutkowski (2007: 329):

- (33) a. mały pancernik olbrzymi  
 small armadillo giant  
 ‘a small *giant armadillo* (i.e., a small representative of *Priodontes maximus*)’  
 b. olbrzymi pancernik mały  
 giant armadillo small  
 ‘a giant *dwarf armadillo* (i.e., a giant representative of *Zaedyus pichiy*)’

While the two NPs in (33a–b) contain the same words, their interpretation is different: (33a) may only refer to a giant armadillo which happens to be small (for this genus), while (33b) refers to a dwarf armadillo which happens to be particularly large (for this genus).

Another aspect of the special status of postnominal APs is that apparently only one such AP may appear in a nominal phrase, while there is no limit on the number of prenominal APs. For example, the attested (34a) with two prenominal APs cannot be transformed into (34b) with two postnominal APs, even though each of these APs alone may occur postnominally.<sup>13</sup>

<sup>12</sup>Such exponence rules are more naturally expressed in L<sub>R</sub>FG, with which the present analysis is meant to be compatible, but which has not been employed here for simplicity.

<sup>13</sup>The subjective judgements reported below assume neutral prosody. (34b) could be saved by a strong prosodic break after *podkładka* ‘placemat’ and the “open-ended list” prosody of the following two APs typical of asyndetic coordination (Büring & Hartmann 2015), with the resulting meaning ‘There was a placemat on the table; and it was plastic, transparent...’.

- (34) a. Na stole leżała plastikowa przezroczysta podkładka.  
 on table was plastic transparent placemat  
 ‘There was a transparent plastic placemat on the table.’  
 b. ?\*Na stole leżała podkładka plastikowa przezroczysta.  
 on table was placemat plastic transparent  
 c. Na stole leżała podkładka {plastikowa / ?przezroczysta}.  
 on table was placemat plastic transparent

Given the possibility of such a postnominal AP and its special classificatory status (argued for at greater length in Rutkowski & Progovac 2005), the NP rule in (23) should be replaced with (35).

- (35) NP → AP\* N (AP) (final version, replaces (23))  
 $\downarrow \in (\uparrow \text{ADJ})$   $\uparrow = \downarrow$   $\downarrow = (\uparrow \text{CLASS})$   
 @CONCORD @CONCORD

Note that the postnominal AP is not treated here as an ordinary adjunct, but rather as a kind of optional classificatory complement (CLASS). Given this treatment, what is special about pronouns such as *coś* ‘something’ is that they idiosyncratically mark their CLASS dependent as @GEN, e.g., in the case of the form *coś* syncretic between NOM and ACC:

- (36) *coś* N  $(\uparrow \text{PRED}) = \text{‘PRO’}$   
 $(\uparrow \text{PHI NUMBER}) = \text{SG}$   
 $(\uparrow \text{PHI GENDER}) = \text{N}$   
 $\neg(\uparrow \text{CASE GEN})$   
 $(\uparrow \text{CLASS}) \rightarrow @\text{GEN}((\uparrow \text{CLASS}))$

The first 4 lines of (36) should be clear; in particular, the possibility of *coś* to expone NOM or ACC but no other case is encoded as a negative constraint forbidding the presence of GEN (cf. (30) above). The last line is an implication saying that, if the CLASS dependent is present, it is marked as (at least) genitive.<sup>14</sup> This lexical entry uses more general parameterized case macros defined in (37)–(42). Given such macros, the non-parameterized macros employed above, such as @NOM or @GEN, may be understood as shorthand notations for corresponding macros with the parameter  $\downarrow$ , e.g., @NOM( $\downarrow$ ) or @GEN( $\downarrow$ ).

- (37) @NOM(X) := (X CASE NOM) = +  
 (38) @ACC(X) := @NOM(X)  
 (X CASE ACC) = +  
 (39) @GEN(X) := @ACC(X)  
 (X CASE GEN) = +  
 (40) @LOC(X) := @GEN(X)  
 (X CASE LOC) = +  
 (41) @DAT(X) := @LOC(X)  
 (X CASE DAT) = +  
 (42) @INS(X) := @DAT(X)  
 (X CASE INS) = +

The net effect of the presence of @CONCORD on the AP classifier in the syntactic rule (35) and the presence of @GEN(( $\uparrow$  CLASS)) in the lexical entry (36) is that both macros apply to the postnominal AP, resulting in the same MACA pattern as in North Slavic NNCs.

<sup>14</sup>Ideally, this constraint should be defined in the grammar just once, for the pronominal root *c(o)*, rather than separately for each pronoun based on this root, but it is not clear to me how to do this either in traditional LFG or in L<sub>R</sub>FG.

Note that the @GEN macro has to be stated in the lexicon, as postnominal APs display the MACA pattern only when they combine with the *c(o)* pronouns, not when they combine with other pronouns (recall table 3) or ordinary nouns. Moreover, while prenominal APs modifying pronouns are rare, when they occur, they always agree with the pronoun, even in NOM or ACC, as in the following examples from the National Corpus of Polish:

- (43) ...czuła to, co czuje teraz: niemożność złapania oddechu, [jakieś coś  
felt.3SG.F this what feels now impossibility catching breath some.ACC something.ACC  
niemiłego]...  
unpleasant.GEN  
'...she felt what she feels now: the inability to catch her breath, something unpleasant...'
- (44) ...a [podobne coś takiego] ma Dagger w Jacku Młocie.  
and similar.ACC something.ACC such.GEN has Dagger in Jacek Młot  
'...and Dagger in Jacek Młot has something similar.' (lit.: '...similar something of-this-kind')

In (43), *jakieś* 'some, some kind of' is an adjectival form agreeing with *coś* 'something' also in the accusative case; it is only the postnominal *niemiłego* 'unpleasant' that occurs in the genitive. Similarly, in (44) the prenominal adjective *podobne* 'similar' agrees with *coś* 'something' in the accusative case, while the adjective *takiego* 'such' occurs in the genitive. Such examples confirm the above analysis in terms of the difference between postnominal adjectival classifiers, which may occur in GEN when relevant pronouns are in NOM or ACC, and prenominal modifiers, which always fully agree with the (pro)noun.

This completes the LFG analysis of MACA patterns. In summary, while some positions are @CONCORD positions and some are @GEN positions, there are some positions in Polish – nominal dependents of numerals, postnominal adjectival dependents of some pronouns – which are specified as both @CONCORD and @GEN. In such cases, given the above encoding of case hierarchy, genitive wins when concord implies nominative or accusative (or genitive), and concord wins otherwise.

What would be unexpected on this analysis is GEN winning not only in NOM and ACC positions, but also in some oblique positions involving a case that is more marked than GEN, e.g., in INS positions. However, apparently exactly this happens in NNCs involving collective numerals.

## 4 Collective numerals

### 4.1 Empirical generalizations

Polish has special numeral lexemes – or, on a minority view, special forms of cardinal numeral lexemes (Saloni 2009) – which are used with certain classes of nouns. The usual generalization found in grammars is that they are used with certain *plurale tantum* nouns, with neuter nouns referring to young animals, and with masculine nouns referring to groups of mixed gender, as well as in various collocations, including with paired parts of the body, especially eyes (e.g., Swan 2002: 200–201). However, the exact distribution of such collective forms in spoken Polish is a matter of some debate, with the consensus that it is much more limited than the prescriptive norm would have it and that collective numerals are slowly disappearing from Polish, with cardinal numerals increasingly used in their place (e.g., Kucała 1976, Zieniukowa 1986, Dmowska & Karpowicz 1993, Karpowicz 1994, Andrzejczuk 2011, Badyda & Warda-Radys 2016, Siuciak 2018, Stefańczyk 2018, Markiewicz 2021).<sup>15</sup> Nevertheless, they are still irreplaceable in some contexts, for example there is no other way to say 'two children' using a numeral than employing a collective numeral, *dwoje dzieci* 'two.COLL.NOM/ACC children.GEN'.<sup>16</sup> Also, where there is optionality

<sup>15</sup>In the opinion of one diachronic linguist specializing in Polish numerals, collective numerals are still present in contemporary Polish only because of considerable prescriptive pressure (Siuciak 2018). Whether it is so or not (one of the reviewers expresses doubts here), they are apparently stronger in Polish than in other Slavic languages (Stefańczyk 2018: 122).

<sup>16</sup>However, the results of a survey among 13–15-year old students reported in Markiewicz (2021) suggest that even here some of the youngest Poles use cardinal numerals.

between cardinal and collective numerals, as in the case of a group of students of mixed gender, collective numerals are still perceived as relatively commonplace.<sup>17</sup> The two paradigms for ‘five students’ – cardinal and collective – are given in table 4. The paradigm for collective NNCs is presented here in a way that is normally presented in grammars, and its accuracy will be disputed below.

Table 4: Polish NNCs with cardinal and collective numerals (prescriptively)

| position | ‘five students’ (cardinal 5) |                       | ‘five students’ (collective 5) |                       |
|----------|------------------------------|-----------------------|--------------------------------|-----------------------|
|          | ‘five.M’                     | ‘students.M’          | ‘five.COLL’                    | ‘students.M’          |
| NOM      | <i>pięciu.ACC</i>            | <i>studentów.GEN</i>  | <i>pięcioro.ACC</i>            | <i>studentów.GEN</i>  |
| ACC      | <i>pięciu.ACC</i>            | <i>studentów.GEN</i>  | <i>pięcioro.ACC</i>            | <i>studentów.GEN</i>  |
| GEN      | <i>pięciu.GEN</i>            | <i>studentów.GEN</i>  | <i>pięciorga.GEN</i>           | <i>studentów.GEN</i>  |
| DAT      | <i>pięciu.DAT</i>            | <i>studentom.DAT</i>  | <i>pięciorgu.DAT</i>           | <i>studentom.DAT</i>  |
| LOC      | <i>pięciu.LOC</i>            | <i>studentach.LOC</i> | <i>pięciorgu.LOC</i>           | <i>studentach.LOC</i> |
| INS      | <i>pięcioma.INS</i>          | <i>studentami.INS</i> | <i>pięciorgiem.INS</i>         | <i>studentów.GEN</i>  |

Polish collective numerals seem to pose a problem for the analyses above, as well as for earlier analyses of NNCs. The reason can be seen in the last column of table 4: unlike in cardinal NNCs, in collective NNCs the noun is in the genitive not only in NOM and ACC (and GEN), but also in INS positions. In terms of previous analyses (mentioned in section 5), which often rely on the direct vs. oblique distinction or structural vs. lexical/inherent distinction, this is unexpected as Polish instrumental is oblique and lexical/inherent, not direct or structural, like nominative and accusative. In terms of the analyses above, this is unexpected because, unlike nominative and accusative, instrumental is *more* marked than genitive, so it should win the competition for exponence.

My reply to this challenge is that the collective paradigm given in grammars and in table 4 is incomplete. The complete descriptive paradigm, together with diacritics indicating their contemporary usage, is given in table 5.

Table 5: Polish NNCs with collective numerals (descriptively)

| position | ‘five students’ (collective 5) |                                                              |
|----------|--------------------------------|--------------------------------------------------------------|
|          | ‘five.COLL’                    | ‘students.M’                                                 |
| NOM      | <i>pięcioro.ACC</i>            | <i>studentów.GEN</i>                                         |
| ACC      | <i>pięcioro.ACC</i>            | <i>studentów.GEN</i>                                         |
| GEN      | <i>pięciorga.GEN</i>           | <i>studentów.GEN</i>                                         |
| DAT      | <i>pięciorgu.DAT</i>           | <i>studentów.GEN</i> <sup>??</sup> or <i>studentom.DAT</i>   |
| LOC      | <i>pięciorgu.LOC</i>           | <i>studentów.GEN</i> <sup>?</sup> or <i>studentach.LOC</i>   |
| INS      | <i>pięciorgiem.INS</i>         | <i>studentów.GEN</i> or <i>studentami.INS</i> <sup>??*</sup> |

The first descriptive observation is that, contrary to prescriptive rules, genitive nouns are possible in all positions, including DAT and, especially, LOC. This has been noted in the few works that attempted to investigate the matter experimentally: with a small survey among 26 adult Poles reported in Dmowska & Karpowicz (1993), accompanied by recommendations in Karpowicz (1994) about how to modify prescriptive rules, with a larger survey involving 200 students of Polish Philology reported in Stefańczyk (2018), and with another survey among 100 middle school (13–15-year old) students reported in Markiewicz

<sup>17</sup>And they are still prescribed as the only possibility by various prescriptive sources and language clinics, e.g., <https://sjp.pwn.pl/poradnia/haslo/Piecioro-studentow-czy-pieciu-studentow;19306.html> (in 2019).

(2021). In all cases the conclusion is the same: GEN nouns should also be considered grammatical in DAT and LOC positions (Karpowicz 1994: 119–120; Stefańczyk 2018: 130; Markiewicz 2021: 101).

While this conclusion has been mostly ignored in grammars and textbooks, one popular textbook on Polish inflection says that in DAT and LOC “genitive forms of nouns are less frequent, but not to the extent that they should be ignored or discredited” (Bańko 2002: 64; translation mine). As this frequency impression has never been supported by any hard statistics, I searched for all occurrences of forms such as *dwojgu* ‘two.COLL.DAT/LOC’ and *pięciorgu* ‘five.COLL.DAT/LOC’ in the 300-million token balanced subcorpus of the National Corpus of Polish (NCP; cf. footnote 7).<sup>18</sup> After removing 10 false positives and repetitions, the 738 remaining results of this query were manually annotated independently by me and by a linguistically-trained student of Cognitive Science at the University of Warsaw.<sup>19</sup> This annotation consisted of 1) marking examples with the noun elided and, in the case of NNCs in LOC positions, with the noun syncretic between GEN and LOC (e.g., the pronoun *nich* ‘they.ACC/GEN/LOC’) – such examples were excluded from further analysis, and 2) disambiguating the numeral forms for DAT or LOC, as the automatically assigned morphosyntactic tags could not be trusted here. Finally, I adjudicated the differences between the two annotators that occurred in 39 (out of 738) results. In effect, 482 occurrences of DAT and LOC NNCs were found with the noun either agreeing in case with the numeral or occurring in GEN; see table 6.

Table 6: Polish NNCs with DAT and LOC collective numerals in the 300M balanced sample of NCP

| position | noun frequencies |     | noun proportions |       |
|----------|------------------|-----|------------------|-------|
|          | agreeing         | GEN | agreeing         | GEN   |
| DAT      | 377              | 24  | 94.0%            | 6.0%  |
| LOC      | 60               | 21  | 74.1%            | 25.9% |

In the case of DAT positions, the noun occurred in GEN – i.e., shunning prescriptive rules – in 6% of the examples, while in the case of LOC positions, it occurred in GEN in as many as 26%. As over 80% of NCP consists of texts written to be published in print (newspapers, books; see Górski & Łaziński 2012: 33), i.e., texts which should follow prescriptive rules, both these numbers are relatively high.

In partial summary, it is clear both from the results of earlier surveys and from the above corpus investigation that the system of Polish allows for genitive nouns in collective NNCs in all case positions, including – contrary to prescriptive rules – DAT and LOC. It is also clear that it allows for agreeing nominal forms in DAT and LOC positions, in line with prescriptive rules. However, the prescriptive position that it does not allow for agreeing INS forms of nouns in INS positions has not been disputed so far. In the remainder of this subsection, I will attempt to cast some doubts also on this claim.

It is true that most Polish native speakers, including the author, usually do not accept and would probably never produce phrases such as *z pięciorgiem studentami* ‘with five.COLL.INS students.INS’. However, there are a few such phrases in the complete NCP, mostly in Usenet posts, involving collective numerals *dwoje* ‘two.COLL’ and *oboje* ‘both.COLL’, e.g.:<sup>20</sup>

- (45) IMHO coś nie tak jest z mamusią / obojgiem rodzicami.  
 IMHO something NEG so is with mummy.INS both.COLL.INS parents.INS  
 ‘IMHO, there is something wrong with the mummy or with both parents.’

Also, Markiewicz (2021: 106–107) reports that, while most middle school students used cardinal numerals instead of the prescribed collective numerals, some (but very few) of those that did use collective numerals “wrongly” put the noun in the instrumental.

<sup>18</sup>The specific query used: [pos="num.\*" & orth=".\*o[rj]gu"].

<sup>19</sup>I am grateful to Aleksandra Wolska for her help at this stage of the process.

<sup>20</sup>There are 7 such examples, plus additional 4 cited in Usenet discussions about grammaticality of such constructions.

It is easy to interpret this scarcity of INS nouns in collective NNCs as editing or production errors, but diachronic developments suggest an alternative explanation. We will first look at the development of *cardinal* numerals, and then we will return to *collective* numerals.<sup>21</sup>

Proto-Indo-European numerals 5 and above were recategorized as mostly feminine nouns in Proto-Slavic, and remained such until the late stages of Old Polish, i.e., until around the 15th century. Similarly to other nouns, they combined with nominal phrases in the genitive, so only case assignment was observed. Around the early 16th century, such semantically numeral nouns started developing the MACA pattern illustrated in table 2, i.e., they started “grammaticalizing” as *cardinal* numerals. Interestingly, this was a long process: numeral-noun case agreement was first observed in DAT positions (becoming the only option around the early 17th century), then in LOC positions (becoming obligatory in the mid-18th century), and only subsequently in INS positions (obligatory since the mid-19th century). See table 7.

Table 7: Development of agreeing forms of NNCs in Polish across centuries; when two dates are given, D1 / D2, agreement started being as frequent as GEN in D1 and took over more or less completely in D2

| numerals   | DAT               | LOC              | INS                                                    |
|------------|-------------------|------------------|--------------------------------------------------------|
| cardinal   | 16th / early 17th | 16th / mid-18th  | early 18th / mid-19th                                  |
| collective | (no data) / 17th  | 17th / (no data) | (infrequently and briefly in late 18th and early 19th) |

Now, *collective* numerals seem to have set out on the same path, but they had a longer way to go, as they started their life in Proto-Slavic as adjectives modifying singular nouns, with meanings such as ‘twofold’, and only such uses existed until the 15th century (Stefańczyk 2018: 121). Such uses were still reported by Michał Abraham Trotz in his 1764 dictionary, e.g., *troie zwycięstwo* ‘threefold victory’ (Trotz 1764: 2336), although they seem to have vanished from actual use by around that time (Stapor 2007: 570; Siuciak 2018: 109). In the meantime, around the 16th century, apparently nominal uses of collective numerals started spreading, which always combined with GEN nouns, as in the relevant column in table 5 (Siuciak 2008: 170–171). Recall that at this stage *cardinal* numerals were already on their way to develop the MACA pattern.

Crucially, the newly formed *collective* numerals followed suit, but with a delay (Siuciak 2008: 171–173). As the frequency of collective numerals is relatively low, it is difficult to give specific times of when a given change occurred; however, only case agreement in collective NNCs is observed in DAT positions already in the 17th century. As for LOC positions, both agreeing and GEN forms of nouns are witnessed with *collective* numerals at that time (even within one publication), even though only agreeing forms are attested with *cardinal* numerals in this period. Finally, in the case of INS positions, there are attested examples of agreeing forms occurring in the 2nd half of the 18th century, and at least one mid-19th century grammar admits such constructions (Łazowski 1848: 140), in line with table 5.<sup>22</sup> However, subsequent and more influential grammars only allow for GEN nouns in instrumental collective NNCs (Małecki 1863: 126).

A convincing interpretation of these developments in Siuciak (2008: 173) is that collective numerals were well under way to follow the development of cardinal numerals and to exhibit exactly the same MACA pattern, but this development was artificially arrested by the intensive codification trends in the 19th century and in the early 20th century (Walczak 2011: 106; Siuciak 2020). This resulted in the freezing of the unstable state where agreement was already dominant in DAT and LOC positions, but where

<sup>21</sup>The following paragraphs draw heavily on Siuciak (2008).

<sup>22</sup>The specific example given there is:

- (i) *pieciorgiem ludzi* (albo *ludźmi*) *nie wiele zrobisz*  
 five.COLL.INS people.GEN or people.INS NEG much do.2SG.FUT  
 ‘you won’t do much with (i.e., having) five people’

GEN assignment was still almost obligatory in INS positions; this way the nascent agreement pattern in INS was eradicated by prescriptivists.

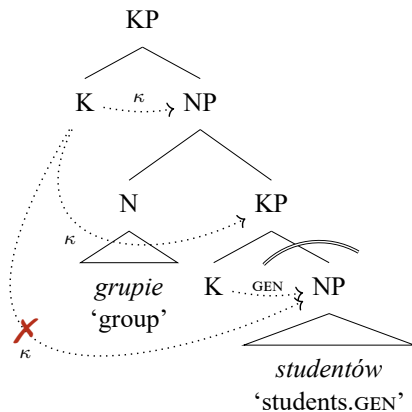
The following subsection demonstrates that this diachrony-driven interpretation of synchronic data is compatible with the analyses proposed above.

## 4.2 Analytical considerations

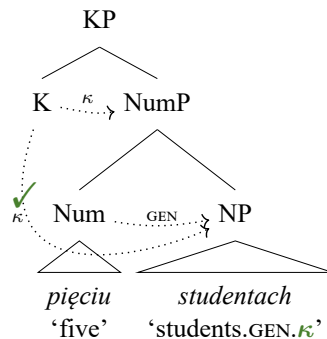
### 4.2.1 Minimalism

As discussed immediately above, cardinal numerals stem from nouns, so – assuming an analysis of numeral phrases as headed by numerals (i.e., as (7) rather than (8)) – a change occurred from a structure like (46) below to a structure like (47). Note that this process involved changing the category from N to Num and changing the complement from KP (i.e., a phase) to NP (not a phase). In the case of collective numerals, this development was arrested at the stage where fully transitioned – but defective due to the prescriptive intervention – collective numerals (as in (49)) co-exist with items that have not undergone the full transition (as in (48)).<sup>23</sup>

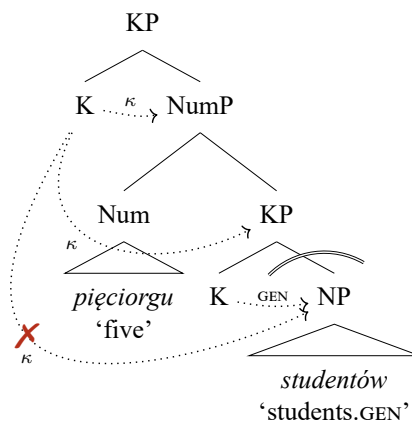
(46) noun-noun ( $\approx$  (6)):



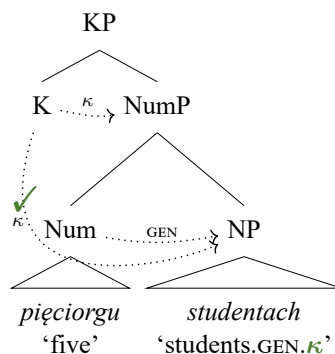
(47) cardinal-noun ( $\approx$  (7)):



(48) nounish collective-noun ( $\sim$  (46)):



(49) cardinalish collective-noun ( $\approx$  (47)):



The partially transitioned collective numerals, illustrated in (48), have changed their category from N to Num, as is clear from the fact that – like prototypical cardinal numerals – they are inherently plural and trigger the default 3SG.N agreement on the verb, as illustrated with the attested (50):

<sup>23</sup> The terminals in the above trees correspond to  $\kappa = \text{LOC}$ , i.e., the forms *grupie* 'group', *pięciu* 'five.CARD', *pięciorgu* 'five.COLL', and *studentach* 'students' may be interpreted as locative. ("May" because – with the exception of *studentach* 'student.PL.M.LOC – they are all syncretic with some other cases.)

- (50) *Te czworo czy pięcioro dzieci mogłoby się przysłużyć państwu.*  
 these.PL four.COLL or five.COLL children.GEN could.3SG.N REFL serve.INF state.DAT  
 ‘These four or five children could be of service to the state.’

The adjectival demonstrative *te* ‘these.NOM/ACC’ agrees with the numeral, and not with the genitive noun; this is clear as *te* is not in the genitive. (Agreement with the noun would also be possible here, but then the form would be the genitive *tych*.) Given that *te* is plural, the numeral must be analysed as plural – this is a standard and generally accepted argument for the inherent plural number of Polish numerals; no singular form would be possible here. However, despite this plurality of the subject, the verb occurs in 3rd person *singular*, which is the default form reflecting the lack of agreement (Dziwirek 1990, Preminger 2014). This is again typical of Polish numerals and it is often explained as reflecting the accusative case of the numeral (Przepiórkowski 1999, 2004, Witkoś et al. 2018, Citko et al. 2018) – an aspect of Polish numerals ignored here.<sup>24</sup>

The above considerations show that collective numerals have successfully changed their category to Num. However, they have not traversed the whole path regarding their complement: they are indeterminate as to whether they still, like nouns, combine with genitive KPs (as in (48)), or whether they are integrated into the noun’s extended projection, like cardinal numerals (as in (49)). In the first case, the complement is realized as GEN, given that KPs are phases, and in the second the nominal complement is marked with the same case  $\kappa$  as the numeral, creating the impression of agreement when  $\kappa$  is at least as marked as GEN. Additionally, given the prescriptivist intervention discussed above, this last option is artificially blocked when  $\kappa = \text{INS}$ .<sup>25</sup>

#### 4.2.2 LFG

While not much hinges on this, I assume that both cardinal and collective numerals have the syntactic category Num but differ in the value of the NUMTYPE feature specified in lexical entries of numerals: CARD or COLL. As discussed immediately above, there are various similarities between cardinal and collective numeral phrases that warrant this treatment, including the fact that they are accusative in subject positions and, consequently, trigger the default third person singular neuter agreement on the verb.

For those speakers who allow for all the possibilities shown in table 5, including agreement in INS, the rule for collective numeral phrases, given in (52), is almost identical to that for cardinal numeral phrases, repeated (with NUMTYPE added) in (51). The only difference is that, in the case of collective numerals, @CONCORD is optional, so the noun may occur in GEN in all positions, even when the numeral bears a case that is more marked than GEN.

- (51) cardinal numerals (modified from (26))

$$\begin{array}{ccc}
 \text{NumP} & \rightarrow & \begin{array}{c} \text{Num} \\ \uparrow = \downarrow \\ (\downarrow \text{NUMTYPE}) =_c \text{CARD} \end{array} & \begin{array}{c} \text{NP} \\ \downarrow = (\uparrow \text{OBJ}) \\ @\text{CONCORD} \\ @\text{GEN} \end{array}
 \end{array}$$

- (52) collective numerals (permissive)

$$\begin{array}{ccc}
 \text{NumP} & \rightarrow & \begin{array}{c} \text{Num} \\ \uparrow = \downarrow \\ (\downarrow \text{NUMTYPE}) =_c \text{COLL} \end{array} & \begin{array}{c} \text{NP} \\ \downarrow = (\uparrow \text{OBJ}) \\ (@\text{CONCORD}) \\ @\text{GEN} \end{array}
 \end{array}$$

<sup>24</sup>Given these considerations, a fuller gloss of *czworo* in (50) would be ‘four.COLL.ACC.PL’, and similarly for *pięcioro* ‘five.COLL.ACC.PL’, with *te* glossed as ‘these.ACC.PL’. (We ignore gender here.)

<sup>25</sup>Technically, this can be accounted for by the assumption that the feature bundle corresponding to an instrumental collective numeral categorially selecting for an NP – i.e., the bundle [Num, COLL+, INS, C:NP] – has no corresponding exponence rule.

## (53) collective numerals (restrictive)

$$\begin{array}{ccc} \text{NumP} & \rightarrow & \begin{array}{c} \text{Num} \\ \uparrow=\downarrow \\ (\downarrow \text{NUMTYPE}) =_c \text{COLL} \end{array} & \begin{array}{c} \text{NP} \\ \downarrow = (\uparrow \text{OBJ}) \\ (@\text{CONCORD} \wedge \neg(\uparrow \text{CASE INS})) \\ @\text{GEN} \end{array} \end{array}$$

However, grammars of most speakers seem to follow the prescriptive rule that forbids concord in *INS*, which leads to the more complex rule in (53), where the optional *@CONCORD* is conditioned on the non-instrumental case of the numeral phrase.

In terms of diachronic change, the specification of dependent NPs evolved from (54a), which is how genitive dependents of nouns may be encoded,<sup>26</sup> to (54c), which shows the encoding of NP dependents of cardinal numerals (cf. (51)), via (54b), which encodes the most permissive specification of NP dependents of collective numerals (cf. (52)).<sup>27</sup>

## (54) nominal dependents of nouns and numerals:

|                                                                                                |                                                                                                                    |                                                                                                                  |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| a. of nouns:                                                                                   | b. of collective numerals:                                                                                         | c. of cardinal numerals:                                                                                         |
| $\begin{array}{c} \text{NP} \\ \downarrow = (\uparrow \text{POSS}) \\ @\text{GEN} \end{array}$ | $\begin{array}{c} \text{NP} \\ \downarrow = (\uparrow \text{OBJ}) \\ @\text{GEN} \\ (@\text{CONCORD}) \end{array}$ | $\begin{array}{c} \text{NP} \\ \downarrow = (\uparrow \text{OBJ}) \\ @\text{GEN} \\ @\text{CONCORD} \end{array}$ |

In summary, collective numerals do not pose a challenge for the analysis of North Slavic NNCs offered in previous sections. On the contrary, once correct empirical generalizations are discovered, it is easy to extend the analysis of cardinal numeral phrases to collective numeral phrases.

## 5 Previous work

### 5.1 LFG

To the best of my knowledge, there is no previous LFG work convincingly explaining MACA patterns in general or NNCs in particular. Neidle (1988), an LFG monograph on Russian case, briefly considers (in §4.1.2) the typical paradigm of Russian numerals (illustrated in table 1 above) and proposes that each apparent numeral lexeme is a conflation of two defective lexemes: nominal (hence the assignment of *GEN* in these positions; only *NOM* and *ACC*) and adjectival (hence the case concord in these positions; *GEN*, *DAT*, *LOC*, *INS*), as shown in table 8.

However, this explanation is very unlikely to be correct, as defective paradigms are usually a matter of some idiosyncrasy, while here all cardinal numerals would actually be pairs of identically complementarily defective lexemes – and not only in Russian, but also in other North Slavic languages.<sup>28</sup>

Another LFG work that attempts to codify the MACA pattern is Przepiórkowski & Patejuk (2012), where the rule (55) (much simplified here) is stated explicitly, and a rule like (56) is assumed.

$$(55) \quad \begin{array}{ccc} \text{NumP} & \rightarrow & \begin{array}{c} \text{Num} \\ (\downarrow \text{ACM}) =_c \text{REC} \\ \uparrow=\downarrow \end{array} & \begin{array}{c} \text{NP} \\ (\downarrow \text{CASE GEN}) =_c + \end{array} \end{array}$$

$$(56) \quad \begin{array}{ccc} \text{NumP} & \rightarrow & \begin{array}{c} \text{Num} \\ (\downarrow \text{ACM}) \neq \text{REC} \\ \uparrow=\downarrow \end{array} & \begin{array}{c} \text{NP} \\ (\downarrow \text{CASE}) = (\uparrow \text{CASE}) \end{array} \end{array}$$

<sup>26</sup>Such a dependent could be specified in an extended version of rule (35) above.

<sup>27</sup>Note that the differences would be even smaller if grammatical functions were drastically reduced in LFG, as advocated by Alsina (1996), Patejuk & Przepiórkowski (2016), Przepiórkowski (2016), and Lowe (2025), as then the equations defining grammatical functions – now *POSS* vs. *OBJ* – might be the same.

<sup>28</sup>Apparently, Neidle (1988) is the sole work in the whole vast literature on Slavic NNCs that assumes such a systematic and complementary defectiveness.

Table 8: Russian numeral-noun constructions according to Neidle (1988)

| position | ‘ f i v e   c h a i r s ’ |                     |                     |                        |
|----------|---------------------------|---------------------|---------------------|------------------------|
|          | 5 is a noun               |                     | 5 is an adjective   |                        |
|          | ‘five’                    | ‘chairs’            | ‘five’              | ‘chairs’               |
| NOM      | <i>pjat’</i> .NOM         | <i>stul’ev</i> .GEN | *                   |                        |
| ACC      | <i>pjat’</i> .ACC         | <i>stul’ev</i> .GEN | *                   |                        |
| GEN      | *                         |                     | <i>pjati</i> .GEN   | <i>stul’ev</i> .GEN    |
| DAT      | *                         |                     | <i>pjati</i> .DAT   | <i>stuljam</i> .DAT    |
| LOC      | *                         |                     | <i>pjati</i> .LOC   | <i>stuljax</i> .LOC    |
| INS      | *                         |                     | <i>pjat’ju</i> .INS | <i>stuljami</i> .M.INS |

These rules assume the “complex case” approach of Dalrymple et al. (2009), according to which [GEN+] on a word means that this word may be interpreted as genitive, while [GEN−] means that it cannot be a genitive form.<sup>29</sup> Additionally, these rules employ the grammatical feature called *accommodability*, abbreviated here to ACM, introduced in Polish structuralist linguistics (Bień & Saloni 1982) to distinguish between numeral forms agreeing with their dependent nouns ([ACM CONGR]) and those requiring the genitive case ([ACM REC]). Hence, the effect of these rules is that [ACM REC] numerals combine with genitive nouns and other numerals combine with nouns agreeing in case.

However, this account does not say anything about the distribution of [ACM REC] on numeral forms. On the assumption that the functional description ( $\uparrow$  ACM) = REC is present on all and only lexical entries for (NOM and) ACC numeral forms, this account ‘works’ in a technical sense, but it is as stipulative as the proposal in Neidle (1988), which relies on highly regular and complementary defectiveness.

By contrast, the LFG account proposed above is free from such lexical stipulations. Instead, it relies on the independently motivated case hierarchy, its LFG encoding based on work within L<sub>R</sub>FG, and a general approach to nominal concord.

## 5.2 Outside LFG

While there is hardly any work on Slavic NNCs in LFG, the derivational literature on this topic is vast – too large to do any justice here, so below I only mention some of the most relevant derivational (and HPSG) works.<sup>30</sup> Most of the work in 1980s and 1990s, of which Babby (1987) and Franks (1994, 1995) are prominent exemplars, crucially relied on the distinction between Deep Structure and Surface Structure assumed within the Government and Binding Theory (GB; Chomsky 1981), but given up in Minimalism. Additionally, that work assumed ad hoc case percolation mechanisms, which the approach to concord proposed in Przepiórkowski (2025) and summarized in section 2 attempts to avoid, as well as a variety of other mechanisms motivated only by the phenomenon of NNCs. One example of such a mechanism is the resolution of conflicts in case assignment on the basis of markedness, with oblique cases (e.g., DAT, INS, etc.) defined as more marked than the genitive of quantification, which in turn is defined as more marked than direct cases (NOM, ACC). While this hierarchy – and this particular aspect of Babby’s (1987) analysis – is broadly compatible with the one assumed here, it was not motivated in Babby (1987) outside of its usefulness to account for Russian NNCs.

Foreshadowing one of the ideas on which the analyses offered in this paper are based, Matushansky (2008: §5.2) proposes that, when essive and translative compete for the exponence of Finnish predicates, translative wins, as it is more marked than essive. In this case, unlike in Babby (1987), markedness

<sup>29</sup>So, for example, [NOM−, ACC+, GEN+, DAT−, LOC−, INS−] indicates that a given form is syncretic (only) between ACC and GEN.

<sup>30</sup>Apart from the broadly generative tradition, there is important dependency work on Russian NNCs by Igor A. Mel’čuk (1985, 1988, 2009), but I do not discuss it here for lack of space.

is defined as in Distributed Morphology, i.e., in terms of featural richness: *essive* expones the feature [PREDICATIVE], while *translative* expones the more marked feature combination [PREDICATIVE, AFFECTED]. Matushansky (2008: §6) also briefly suggests that this approach to case exponence may explain the MACA pattern of Russian NNCs, but she does not provide a worked out analysis.<sup>31</sup>

Another derivational analysis which bears certain resemblance to the analyses proposed here is that of Witkoś et al. (2018). Not only does it assume Caha's (2009) case hierarchy, but it also adopts specific Nanosyntax mechanisms and representations. However, the part of the analysis aiming to explain the difference between agreement and case assignment patterns is rather complex, it is based on a number of unusual assumptions (see, e.g., Witkoś et al. 2018: 64, fn. 35), and it leaves many questions open, so a more detailed discussion of this approach would require considerable space, which I lack here.

Finally, an HPSG analysis of Polish NNCs inspired by some GB ideas, especially, the distinction between structural and lexical case assignment, is offered in Przepiórkowski (1999) (and related publications). On this analysis, numerals in structural positions require genitive dependents and those in lexical positions require agreeing dependents. Given the power of HPSG, this statement can be made once in the grammar, rather than – missing generalizations – in a large number of lexical entries, as in the LFG analyses discussed in section 5.1. Nevertheless, apart from being more stipulative than the analyses offered above (why should there be such a difference between structural and lexical positions?), it faces a problem related to arguments of prepositions.

Morphological realization of cases assigned structurally depends on particular syntactic configurations. For example, in Polish, structurally-cased nominal subjects of verbs are exponed as nominative and structurally-cased objects as accusative (in the absence of negation). It has been similarly proposed (e.g., in Franks 1995) that, unless a preposition assigns a specific lexical case (e.g., LOC or INS) to its argument, this argument bears a structural case, realized as accusative. However, there are also some (very few) prepositions in Polish that combine with nominative NPs, e.g., *via* 'via, through'. If accusative is the structural case of arguments of prepositions, then such nominative NPs must bear the *lexical* nominative. But numeral phrases behave in such nominative positions as in "structural" positions, not as in "lexical" positions: the noun is in the genitive, and not in the nominative expected on the analysis in Przepiórkowski (1999) if nominative assigned by prepositions were lexical.<sup>32</sup> This is illustrated by the following attested example, ungrammatical when the GEN form *balkonów* is changed to the NOM/ACC form *balkony*:<sup>33</sup>

- (57) *nocna eskapada Julii [via siedem {balkonów / \*balkony}] do pokoju żonatego*  
 nightly escapade Julia.GEN via seven balconies.GEN/\*{NOM/ACC} to room married.GEN  
*mężczyzny*  
 man.GEN  
 'Julia's nighttime escapade via seven balconies to a married man's room'

The analyses proposed in this paper, which rely on the markedness of morphological cases rather than on the structural/lexical distinction, are free from this problem.

## 6 Conclusion

I believe that the above account of MACA patterns is considerably less stipulative than previous analyses. In the case of NNCs, it does not rely on any lexical stipulations, but it does rely on a lexical stipulation where it seems unavoidable: in the case of the 'something interesting' idiosyncrasy. It also relies on an independently motivated case hierarchy, rather than on a stipulated hierarchy or on the structural/lexical

<sup>31</sup>I became aware of this proceedings paper on predication only after developing the account presented in Przepiórkowski (2025). Note that also Ionin & Matushansky (2018: 185) seem to assume such an *expones the most marked case* principle in their account of Russian paucals.

<sup>32</sup>Conversely, if ad-prepositional nominative were taken to be structural and ad-prepositional accusative lexical, then ad-numeral nouns in such ad-prepositional accusative positions would be wrongly expected to occur in the accusative rather than in the genitive.

<sup>33</sup><https://ruj.uj.edu.pl/bitstreams/b17150a7-e61b-4cc2-b343-dac0868322f6/download>

dichotomy and the timing of case assignment, as in the case of the best known derivational analyses. Finally, each of the two implementations of the account relies on general mechanisms independently employed in the respective framework: Agree in Minimalism and a particular encoding of case proposed in  $L_{\text{RFG}}$ .

On the empirical side, I adduced new data showing that collective numerals – usually avoided in discussions of Slavic numerals, probably due to the problems they seem to pose for most analyses – are not a problem for the proposed account, once the correct empirical generalizations are found. Hopefully, these empirical considerations will make it possible to include collective numerals in future analyses – before this class of numerals disappears from Slavic altogether.

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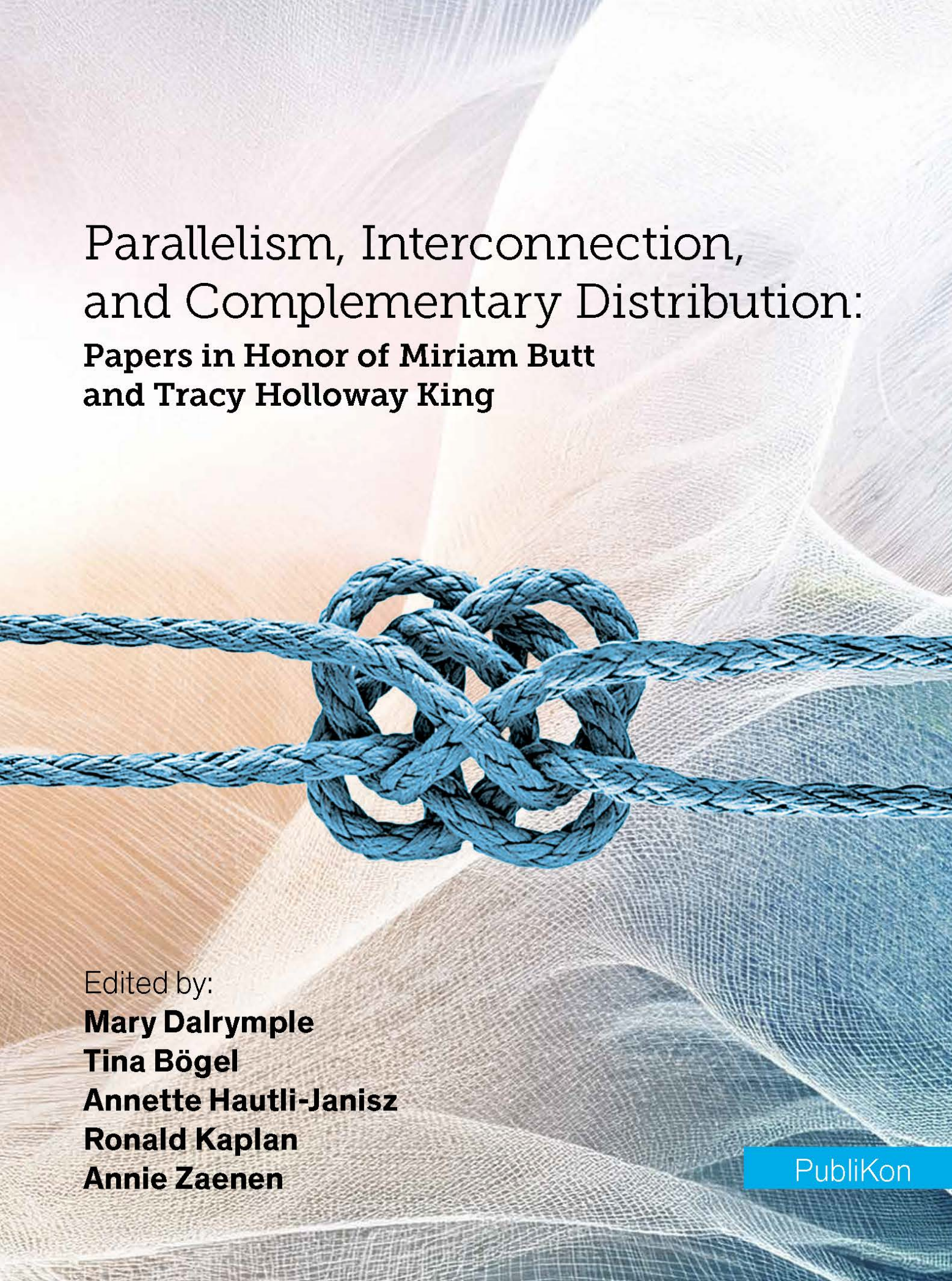
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# Parallelism, Interconnection, and Complementary Distribution: Papers in Honor of Miriam Butt and Tracy Holloway King

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